

Section : Chemistry

Q.1 The rate constant for a zero order reaction was found to be $1.0 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$.
How long will 1.0 mol of the reactant take to reduce to 0.5 mol in one liter flask?

1. 500 s
2. 1000 s
3. 2000 s
4. 5000 s

Options 1. 1
2. 2
3. 3
4. 4

Q.2 Match List-I with List-II

List-I (Name of Reaction) List-II (Reagents)

- | | |
|-----------------------------|---|
| (A) Stephens reaction | (I) $\text{CrO}_2\text{Cl}_2/\text{CS}_2$ |
| (B) Rosenmund reduction | (II) Acyl chloride and anhyd. AlCl_3 |
| (C) Etard reaction | (III) SnCl_2 , HCl |
| (D) Friedel Craft acylation | (IV) $\text{H}_2/\text{Pd}-\text{BaSO}_4$ |

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
2. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options 1. 1
2. 2
3. 3
4. 4

Q.3 If the rate constant for the first order reaction is $2 \times 10^{-5} \text{ s}^{-1}$, then the half-life will be

1. 577.5 min
2. 677.5 min
3. 277.5 min
4. 177.5 min

Options 1. 1
2. 2
3. 3
4. 4

Q.4 Gluconic acid is obtained when

1. Sucrose is boiled with dil HCl or H_2SO_4 in alcoholic solution.
2. Glucose is treated with a mild oxidizing agent like bromine water.
3. Upon hydrolysis of starch with dilute H_2SO_4 at 393 K under pressure.
4. Glucose is prolonged heated with hydrogen iodide.

Options 1. 1
2. 2
3. 3
4. 4

Q.5 Match the statement of set of orbitals given in List-1 with orbitals and energy separation given in List-II

List-I (Statements)

List-II (Orbitals and energy separation)

- (A) t_{2g} set of orbitals in octahedral crystal field and energy separation from barycentre (I) $d_{x^2-y^2}$, d_{z^2} and $3/5\Delta_t$

List-I (Statements)

- (B) e_g set of orbitals in octahedral crystal field and energy separation from barycentre (II) d_{xy}, d_{xz}, d_{yz} and $2/5\Delta_t$
- (C) t_2 set of orbitals in tetrahedral crystal field and energy separation from barycentre (III) $d_{x^2-y^2}, d_{z^2}$ and $3/5\Delta_0$
- (D) e set of orbitals in tetrahedral crystal field and energy separation from barycentre (IV) d_{xy}, d_{xz}, d_{yz} and $2/5\Delta_0$

List-II (Orbitals and energy separation)

Choose the \textbf{correct} answer from the options given below:

- (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Which of the following postulates of Werner theory is incorrect?

- In coordination compounds metals show two types of linkages (valences) - primary and secondary.
- The primary valences are normally ionisable and are satisfied by negative ions.
- The secondary valence is equal to the oxidation number of metal and is fixed for a metal. (Incorrect)
- The ions/groups bound by the secondary linkages to the metal have characteristic spatial arrangements corresponding to different coordination numbers.

Options 1. 1

2. 2

3. 3

4. 4

Q.7 The correct statements regarding order and molecularity of the reaction are:

- The order of a reaction can be zero or an integer.
- The molecularity of a reaction can be zero or an integer.
- The molecularity cannot be zero or fraction.
- The order of a reaction is decided by the slowest step of the complex reaction.

Choose the \textbf{correct} answer from the options given below:

- (A), (B) and (C) only
- (B), (C) and (D) only
- (A), (B) and (D) only
- (A), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.8 In a primary dry cell, choose the \textbf{incorrect} statement.

- Mn is oxidized from +4 to +6 oxidation state.
- Zinc container acts as anode.
- Mn is reduced from +4 to +3 oxidation state.
- Ammonia forms a complex with Zn.

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Ascorbic Acid is

- Vitamin C
- Vitamin D
- Vitamin B
- Vitamin K

Options 1. 1

2. 2

3. 3

4. 4

Q.10 The compound which produce turbidity immediately with Lucas reagent is

1. $(\text{CH}_3)_3\text{C-OH}$

2. $(\text{CH}_3)_2\text{CH-OH}$

3. $\text{C}_6\text{H}_5\text{OH}$

4. $\text{CH}_3\text{CH}_2\text{-OH}$

Options 1. 1

2. 2

3. 3

4. 4

Q.11 The freezing point of an aqueous solution containing 23 g of $\text{C}_2\text{H}_5\text{OH}$ in 1000 g of water, is

(Given K_f for water = $1.86 \text{ K kg mol}^{-1}$)

1. -0.69°C

2. -0.93°C

3. 0.93°C

4. 0.69°C

Options 1. 1

2. 2

3. 3

4. 4

Q.12 The decreasing order of basicity of the following amines in aqueous solution is

(A) Ethylamine

(B) N-Ethylethanamine

(C) N,N-Diethylethanamine

(D) Benzamine

Choose the correct answer from the options given below:

1. (B), (A), (C), (D)

2. (B), (C), (A), (D)

3. (A), (C), (B), (D)

4. (B), (C), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.13 The product (Y) in the benzylation of aniline is



1. $\text{C}_6\text{H}_5\text{NHC}_6\text{H}_5$

2. $2\text{C}_6\text{H}_5\text{NH}_2$

3. $\text{C}_6\text{H}_5\text{NHCOC}_6\text{H}_5$

4. $\text{C}_6\text{H}_5\text{NH}_2\text{COC}_6\text{H}_5$

Options 1. 1

2. 2

3. 3

4. 4

Q.14 The product, when an alcoholic solution of sucrose is boiled with dilute hydrochloric acid or sulphuric acid is/are

1. Glucose only

2. Fructose only

3. 50% glucose and 50% fructose

4. 75% glucose and 25% fructose

Options 1. 1

2. 2
3. 3
4. 4

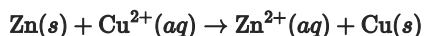
Q.15 Choose the \textbf{correct} statement from the options given below:

1. The solution of benzene and toluene is an example of an ideal solution.
2. For an ideal solution, $\Delta H_{mix} > 0$.
3. Ideal solutions do not obey Raoult's law.
4. The solution of water and alcohol is an example of an ideal solution.

Options

1. 1
2. 2
3. 3
4. 4

Q.16 Select the correct relation between the electrochemical cell and Gibbs free energy for the following reaction



1. $\Delta_r G = -4FE_{cell}$
2. $\Delta_r G = -2FE_{cell}$
3. $\Delta_r G = -1FE_{cell}$
4. $\Delta_r G = 2FE_{cell}$

Options

1. 1
2. 2
3. 3
4. 4

Q.17 Potassium tetrahydroxyzincate(II) can be represented by which of the following formula?

1. $\text{K}_3[\text{Zn}(\text{OH})_4]$
2. $\text{K}_2[\text{Zn}(\text{OH})_4]$
3. $[\text{K}_2(\text{OH})_4]\text{Zn}$
4. $[\text{K}(\text{OH})_4]_2\text{Zn}$

Options

1. 1
2. 2
3. 3
4. 4

Q.18 The examples of disaccharides are

- (A) Sucrose
- (B) Maltose
- (C) Lactose
- (D) Cellulose

Choose the \textbf{correct} answer from the options given below:

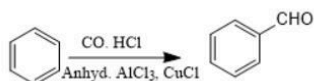
1. (A), (B) and (C) only
2. (A), (B) and (D) only
3. (B), (C) and (D) only
4. (A), (B), (C) and (D)

Options

1. 1
2. 2
3. 3
4. 4

Q.19

The following reaction of benzene with carbon monoxide and hydrogen chloride in the presence of aluminum trichloride to give benzaldehyde is known as



1. Kolbe's reaction
2. Gatterman-Koch reaction
3. Williamson synthesis
4. Friedel-Craft reaction

Options 1. 1

2. 2

3. 3

4. 4

Q.20 The amount of urea required to make a 500 mL solution of 20%(w/w) concentration is

1. 20 g

2. 50 g

3. 100 g

4. 10 g

Options 1. 1

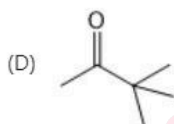
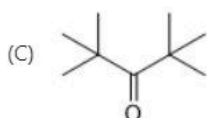
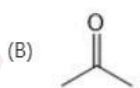
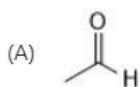
2. 2

3. 3

4. 4

Q.21

Arrange the following compounds in increasing order of their reactivity towards HCN.



Choose the **correct** answer from the options given below:

1. (A), (B), (C), (D)

2. (C), (D), (B), (A)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Choose the **correct** statement from the following:

1. $\text{K}_4[\text{Fe}(\text{CN})_6]$ is a double salt formed by the combination of Fe^{2+} and CN^- ions in stoichiometric ratio

2. $K_4[Fe(CN)_6]$ is a double salt and dissociate into simple Fe^{2+} and CN^- ions completely when dissolved in water
3. $KAl(SO_4)_2 \cdot 12H_2O$ is a coordination complex in which SO_4^{2-} ligand is coordinated to Al
4. $KAl(SO_4)_2 \cdot 12H_2O$ is a double salt and dissociate into simple ions completely when dissolved in water

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Osmotic pressure of 0.25 M aqueous solution of urea at 300 K is

1. 61.50 bar

2. 6.225 bar

3. 615.15 bar

4. 6.15 bar

Options 1. 1

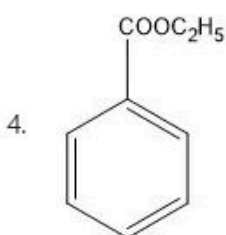
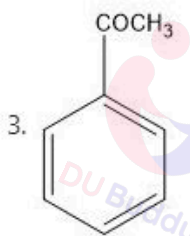
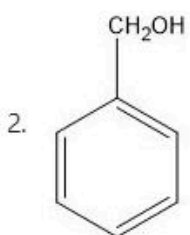
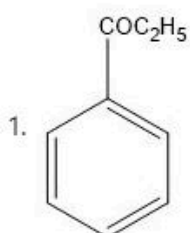
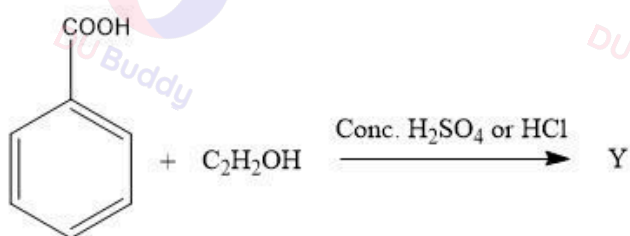
2. 2

3. 3

4. 4

Q.24

Major product Y in the following reaction is



Options 1. 1

2. 2

3. 3

4. 4

Q.25 The amount of urea required to make a 500 mL solution of 0.25 molarity is
(Molar mass of Urea = 60 g mol^{-1})

1. 7.5 g

2. 10 g

3. 60 g

4. 30 g

Options 1. 1

2. 2

3. 3

4. 4

Q.26 According to the electrochemical series, which of the following statements is **incorrect**?

1. Fluorine has the highest standard electrode potential.

2. Fluorine gas has a strong tendency to get reduced.

3. Fluorine gas is a strong oxidizing agent.

4. Fluorine ion is a strong oxidizing agent.

Options 1. 1

2. 2

3. 3

4. 4

Q.27 The molecule having largest C – X bond length is

1. CH_3F

2. CH_3I

3. CH_3Cl

4. CH_3Br

Options 1. 1

2. 2

3. 3

4. 4

Q.28 Match List-I with List-II

List-I

List-II

Order of the reaction

Example

(A) Third order

Extra close brace or missing open brace

(B) First order

Extra close brace or missing open brace

(C) One and half order

Extra close brace or missing open brace

(D) Pseudo first order

Extra close brace or missing open brace

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.29 The reaction of aliphatic and aromatic primary amines with chloroform in the presence of ethanolic potassium hydroxide to produce compound with foul smell, is known as

1. Rosenmund reaction

2. Carbylamine reaction
3. Fittig reaction
4. Etard reaction

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Match List-I with List-II

List-I (Structure of Protein) List-II (Function)

- | | |
|--------------------------|---|
| (A) Primary Structure | (I) Gives rise to Fibrous and Globular Proteins |
| (B) Secondary structure | (II) Sequence of amino acid |
| (C) Tertiary Structure | (III) shape of polypeptide chain |
| (D) Quaternary Structure | (IV) Spatial arrangement of sub-units |

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Out of the following compounds, the chiral molecules are

- (A) Butan-2-ol
- (B) Propane-2-ol
- (C) 2-Chlorobutane
- (D) 2-Bromopropanoic acid

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (A), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Match List-I with List-II

List-I (Property)

List-II (Symbol)

- | | |
|--------------------------|---|
| (A) Resistance | Extra close brace or missing open brace |
| (B) Specific resistance | Extra close brace or missing open brace |
| (C) molar conductivity | Extra close brace or missing open brace |
| (D) specific conductance | Extra close brace or missing open brace |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.33 The reaction which can be used to prepare aryl halides is

1. Aldol reaction
2. Kolbe's reaction

3. Sandmeyer's reaction

4. Reimer-Tiemann reaction

Options 1. 1

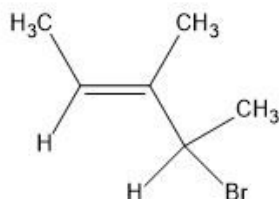
2. 2

3. 3

4. 4

Q.34

The correct IUPAC name of the following compound is



1. 4-Bromo-2-methylpent-2-ene

2. 2-Bromo-3-methylpent-2-ene

3. 2-Bromo-2-methylpent-2-ene

4. 4-Bromo-3-methylpent-2-ene

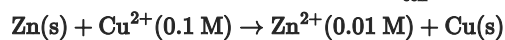
Options 1. 1

2. 2

3. 3

4. 4

Q.35 If the standard electrode potential, $E_{\text{cell}}^{\circ} = 1.10 \text{ V}$, then the E_{cell} for the following reaction at 298 K is



1. 1.1295 V

2. 1.0705 V

3. 1.1 V

4. - 1.1295 V

Options 1. 1

2. 2

3. 3

4. 4

Q.36 In the electrophilic substitution reactions of aniline, the $-\text{NH}_2$ group acts

1. As ortho directing group only

2. As para directing group only

3. As both Ortho and para directing group

4. As meta directing group only

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Arrange the following in the increasing order of rate of the second order reaction

Reactants $\rightarrow$ products

(A) Reaction at 298 K

(B) Reaction at 298 K with catalyst

(C) Reaction at 500 K with catalyst and double the initial concentration of reactant

(D) Reaction at 500 K with double the initial concentration of reactant

Choose the correct answer from the options given below:

1. (A), (B), (D), (C)

2. (A), (B), (C), (D)

3. (D), (B), (C), (A)

4. (A), (C), (D), (B)

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Arrange the following ligands in the increasing order of their field strength:

(A) en

(B) edta^{4-}

(C) S^{2-}

(D) OH^-

Choose the correct answer from the options given below:

1. (C), (D), (B), (A)

2. (A), (B), (D), (C)

3. (A), (D), (B), (C)

4. (C), (B), (D), (A)

Options 1. 1

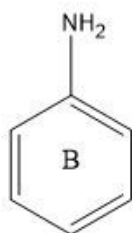
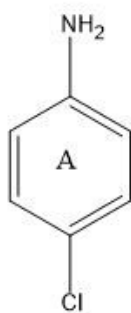
2. 2

3. 3

4. 4

Q.39

Identify the most acidic compound



1. (A)

2. (B)

3. (C)

4. (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.40 Correct order of the following acids in order of their increasing acidic strength is

(A) CCl_3COOH

(B) ClCH_2COOH

(C) CH_3COOH

(D) CHCl_2COOH

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (C), (B), (D), (A)

3. (B), (A), (D), (C)

4. (A), (B), (D), (C)

Options 1. 1

2. 2

3. 3

Q.41 Read the passage carefully and answer the questions

Ethers with the general formula $C_nH_{2n+2}O$ and functional group $C - O - C$ may be divided as symmetrical and unsymmetrical. Lower symmetrical ether may be obtained by the dehydration of alcohols at low temperatures in the presence of protic acids, such as H_2SO_4 and H_3PO_4 whereas at high temperature intramolecular dehydration takes place.

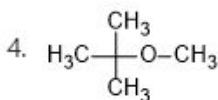
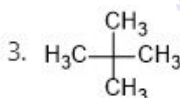
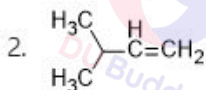
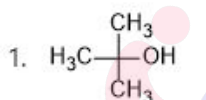
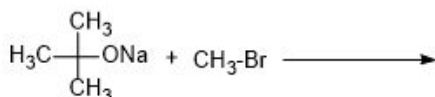
Formation of ether from primary alcohols is also accompanied by the formation of small amount of alkenes.

Due to the steric hinderance, secondary and tertiary alcohols undergo elimination reaction rather than substitution reaction yielding alkene as major product. Ethers can also be obtained by heating alkyl halides with sodium or potassium alkoxides.

Substituted Ethers (secondary & tertiary) can also be prepared using this method.

If primary halides are replaced with secondary and tertiary halides, alkenes are the major products.

Major product in the following reaction is



Options

1. 1
2. 2
3. 3
4. 4

Q.42 Read the passage carefully and answer the questions

Ethers with the general formula $C_nH_{2n+2}O$ and functional group $C - O - C$ may be divided as symmetrical and unsymmetrical. Lower symmetrical ether may be obtained by the dehydration of alcohols at low temperatures in the presence of protic acids, such as H_2SO_4 and H_3PO_4 whereas at high temperature intramolecular dehydration takes place.

Formation of ether from primary alcohols is also accompanied by the formation of small amount of alkenes.

Due to the steric hinderance, secondary and tertiary alcohols undergo elimination reaction rather than substitution reaction yielding alkene as major product. Ethers can also be obtained by heating alkyl halides with sodium or potassium alkoxides.

Substituted Ethers (secondary & tertiary) can also be prepared using this method.

If primary halides are replaced with secondary and tertiary halides, alkenes are the major products.

The major product in the following reaction is $\text{ROH} + \text{HOR} \xrightarrow{H_2SO_4, 140^\circ\text{C}}$

1. Alkenes
2. Ethers
3. Carboxylic acids
4. Aldehydes

Options

1. 1
2. 2
3. 3
4. 4

Q.43 Read the passage carefully and answer the questions

Ethers with the general formula $C_nH_{2n+2}O$ and functional group $C - O - C$ may be divided as symmetrical and unsymmetrical. Lower symmetrical ether may be obtained by the dehydration of alcohols at low temperatures in the presence of protic acids, such as H_2SO_4 and H_3PO_4 whereas at high temperature intramolecular dehydration takes place.

If primary halides are replaced with secondary and tertiary halides, alkenes are the major products.

$$\text{H}_3\text{C}-\underset{\text{OH}}{\overset{\text{H}}{\text{C}}}-\text{CH}_3 \xrightarrow[140^\circ\text{C}]{\text{Conc. H}_2\text{SO}_4} \underset{\text{A}}{\text{H}_3\text{C}-\underset{\text{CH}_3}{\overset{\text{H}}{\text{C}}}-\text{O}-\underset{\text{CH}_3}{\overset{\text{H}}{\text{C}}}-\text{CH}_3} \text{ or } \underset{\text{B}}{\text{H}_3\text{C}-\overset{\text{H}_2}{\text{C}}-\text{O}-\overset{\text{H}_2}{\text{C}}-\text{CH}_3} \text{ or } \underset{\text{C}}{\text{H}_3\text{C}-\underset{\text{H}}{\text{C}}=\text{CH}_2} \text{ or } \underset{\text{D}}{\begin{array}{c} \text{H}_3\text{C} \\ | \\ \text{H}_3\text{C}-\text{C}-\text{O}-\text{C}-\text{CH}_3 \\ | \quad | \\ \text{H}_3\text{C} \quad \text{H}_3\text{C} \end{array}}$$

2. 2
3. 3
4. 4

If primary halides are replaced with secondary and tertiary halides, alkenes are the major products.

$$\text{H}_3\text{C}-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{Br} + \text{CH}_3\text{-ONa} \longrightarrow$$

2. 2
3. 3
4. 4

Ethers with the general formula $C_nH_{2n+2}O$ and functional group $C - O - C$ may be divided as symmetrical and unsymmetrical.

Lower symmetrical ether may be obtained by the dehydration of alcohols at low temperatures in the presence of protic acids, such as H_2SO_4 and H_3PO_4 whereas at high temperature intramolecular dehydration takes place.

Formation of ether from primary alcohols is also accompanied by the formation of small amount of alkenes.

Due to the steric hinderance, secondary and tertiary alcohols undergo elimination reaction rather than substitution reaction yielding alkene as major product. Ethers can also be obtained by heating alkyl halides with sodium or potassium alkoxides.

Substituted Ethers (secondary & tertiary) can also be prepared using this method.

If primary halides are replaced with secondary and tertiary halides, alkenes are the major products.

The compound with the lowest boiling point is

1. Alcohols
2. Aldehydes
3. Ethers
4. Ketones

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Read the passage carefully and answer the Questions

KMnO_4 is prepared by the fusion of MnO_2 with an alkali metal hydroxide and an oxidizing agent like KNO_3 to give a dark-green manganate ion which disproportionate to give permanganate as follows.



On heating KMnO_4 decomposes at 513 K to give K_2MnO_4 . Permanganate ion is tetrahedral and diamagnetic.

Acidified KMnO_4 acts a strong oxidizing agent which oxidizes oxalic acid, ferrous ions, nitrite ion and iodides.

HCl is not used to make the medium acidic in oxidation reaction of KMnO_4 in acidic medium, because....

1. KMnO_4 act as reducing agent in the presence of HCl
2. KMnO_4 oxidizes HCl into Cl_2 which is also an oxidizing agent
3. KMnO_4 is a weaker oxidizing agent than HCl
4. both HCl and KMnO_4 act as oxidizing agent

Options 1. 1

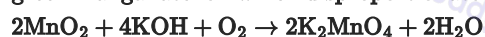
2. 2

3. 3

4. 4

Q.47 Read the passage carefully and answer the Questions

KMnO_4 is prepared by the fusion of MnO_2 with an alkali metal hydroxide and an oxidizing agent like KNO_3 to give a dark-green manganate ion which disproportionate to give permanganate as follows.



On heating KMnO_4 decomposes at 513 K to give K_2MnO_4 . Permanganate ion is tetrahedral and diamagnetic.

Acidified KMnO_4 acts a strong oxidizing agent which oxidizes oxalic acid, ferrous ions, nitrite ion and iodides.

Oxidation states of Mn in MnO_2 , KMnO_4 and K_2MnO_4 are

1. + 4, + 7, and + 7 respectively
2. + 6, + 7, and + 4 respectively
3. + 4, + 6, and + 7 respectively
4. + 4, + 7, and + 6 respectively

Options 1. 1

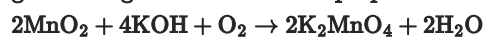
2. 2

3. 3

4. 4

Q.48 Read the passage carefully and answer the Questions

KMnO_4 is prepared by the fusion of MnO_2 with an alkali metal hydroxide and an oxidizing agent like KNO_3 to give a dark-green manganate ion which disproportionate to give permanganate as follows.



On heating KMnO_4 decomposes at 513 K to give K_2MnO_4 . Permanganate ion is tetrahedral and diamagnetic.

Acidified KMnO_4 acts a strong oxidizing agent which oxidizes oxalic acid, ferrous ions, nitrite ion and iodides.

Equivalent weight of KMnO_4 in its oxidising reactions in acidic medium is taken as

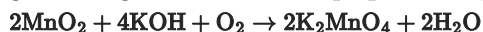
1. One-third of its molecular weight
2. One-fifth of its molecular weight
3. half of its molecular weight
4. Equal to its molecular weight

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the Questions

KMnO_4 is prepared by the fusion of MnO_2 with an alkali metal hydroxide and an oxidizing agent like KNO_3 to give a dark-green manganate ion which disproportionate to give permanganate as follows.



On heating KMnO_4 decomposes at 513 K to give K_2MnO_4 . Permanganate ion is tetrahedral and diamagnetic.

Acidified KMnO_4 acts a strong oxidizing agent which oxidizes oxalic acid, ferrous ions, nitrite ion and iodides.

Reaction of iodide ion (I^-) with KMnO_4 will.....

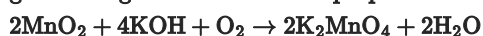
1. liberate I_2 in acidic solution
2. produce IO_3^- in acidic solution
3. liberate I_2 in neutral or mild alkaline solution
4. produce I_2O_4 in neutral or mild alkaline solution

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the Questions

KMnO_4 is prepared by the fusion of MnO_2 with an alkali metal hydroxide and an oxidizing agent like KNO_3 to give a dark-green manganate ion which disproportionate to give permanganate as follows.



On heating KMnO_4 decomposes at 513 K to give K_2MnO_4 . Permanganate ion is tetrahedral and diamagnetic.

Acidified KMnO_4 acts a strong oxidizing agent which oxidizes oxalic acid, ferrous ions, nitrite ion and iodides.

When one mole of KMnO_4 is used for the oxidation of oxalic acid, how many moles of CO_2 will be produced?

1. 5
2. 10
3. $5/2$
4. 2

Options 1. 1

2. 2
3. 3
4. 4